BeeWell

"Your Hive Of Wellness"

Project Report

1. Introduction 💡

Modern life often leads to unhealthy habits, irregular sleep, sedentary routines, and poor diet choices. Many people lack clarity on how these habits affect their long-term health.

BeeWell is an AI/ML-powered wellness dashboard that provides users with:

A predicted overall health score
Lungs and liver risk assessments
Life expectancy projections
Frequency-based analysis of habits
Personalized lifestyle recommendations

The project combines machine learning with interactive visualization to make health insights accessible, actionable, and fun.

2. Problem Statement 📜

People struggle to quantify the impact of their daily habits.
Generic health apps often overwhelm users with raw data or fail to show long-term consequences.
There is a need for a simple, interactive, and engaging tool to help users make informed lifestyle choices.

BeeWell solves this by:✅

Converting lifestyle inputs into a predictive health score
Highlighting organ-specific risks
Showing frequency-based lifestyle impact
Providing practical recommendations

3. Objectives📌

- Predict overall health score based on biometric and lifestyle data
- Assess lungs and liver risk based on smoking and alcohol intake
- Estimate life expectancy influenced by daily habits
- Provide interactive visualizations for better understanding
- Generate actionable recommendations for a healthier lifestyle

4. Dataset

- Synthetic dataset of 500 users created for training the ML model
- Features: Age, Weight, Height, BMI, Cigarettes, Alcohol, Distance Walked, Sleep Hours, Calories, Caffeine
- Health Score calculated using a weighted formula based on:

Smoking and alcohol
Exercise (distance walked)
Sleep hours
Calories and caffeine
BMI and age

5. Methodology

5.1 Data Preprocessing

BMI calculated as:
BMI = weight / (height/100)^2
Features normalized using MinMaxScaler for machine learning
Synthetic dataset ensured diversity and variability in habits

5.2 Machine Learning

Random Forest Regressor trained to predict overall health score
Input features: Age, Weight, Height, BMI, Cigarettes, Alcohol, Distance, Sleep, Calories, Caffeine
Model tuned to capture complex relationships between lifestyle factors and health

5.3 Frequency-Based Analysis

Health predictions adjusted for habit frequency:
Daily
Weekly
Occasionally
Life expectancy adjusted based on cumulative lifestyle risk

6. Implementation📕
Developed using Python in a Jupyter Notebook / Google Colab
Key libraries: pandas, numpy, scikit-learn, matplotlib, ipywidgets
Interactive sliders for user inputs
Real-time health projection tables and graphical visualizations

8. Challenges🏹
Ensuring realistic predictions with synthetic data
Balancing medical accuracy with simplicity and interactivity
Designing recommendations that are actionable yet concise
Making interactive widgets responsive and user-friendly

9. Results🎗️
Accurately predicts health scores for various input scenarios
Frequency-based analysis shows impact of habit changes
Motivational notes and recommendations improve engagement
Interactive visualizations make complex health data understandable

10. Learning Outcomes✅
Practical experience in AI/ML model development
Understanding feature scaling, regression, and synthetic datasets
Creating interactive dashboards with Python libraries
Designing user-centered health applications

11. Conclusion✅

BeeWell is a comprehensive health companion, combining machine learning, data visualization, and interactive design. By turning daily habits into predictive health insights, it empowers users to make informed lifestyle choices.

Its frequency-based projections, organ risk assessments, and personalized recommendations make health tracking practical, actionable, and fun. This project demonstrates the power of AI in personal wellness and provides a strong foundation for future enhancements, such as integrating real-world datasets, mental health indicators, and diet guidance.